Six Configurable FAI Dimensions as Substrate Content

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Parent claim: D0.05 (Paper 3 Claim 5)

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Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in Inter-Self Coordination via Shared Substrate / Full Aspect Integration (Li, April 2026), the third paper in the CKS theory series following Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems (Li, April 2026) and The Instinct/Reasoning Separation Outside the Model (Li, April 2026).

Abstract

Paper 3 Claim 5 commits that every configurable dimension of the Full Aspect Integration (FAI) protocol is human-governed and lives as authored substrate content under joint authority of participating Selves' governance. This note, D1.22, opens the Claim 5 sub-commitment set by enumerating and formalizing the six specific configurable dimensions: (1) sharing scope, (2) cardinality, (3) persistence policy, (4) cooperation/competition variant, (5) provenance carry-over depth at the perimeter, and (6) provenance preservation on internalization. Each dimension is independently claimable as prior art and is subject to Paper 1 Claim 3's three rights—inspect, modify, override—without requiring system or infrastructure permission. Together the six constitute the complete FAI configuration territory at the event level. An operational test is provided. D1.23 through D1.25 cover the recursive, cross-claim, and joint-authority properties that complete the Claim 5 sub-commitment set.

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1. Derivation chain and purpose of this note

Paper 1 Claim 3 commits that humans hold three rights over all substrate content and orchestration rules at all times: the right to inspect any such content, the right to modify it, and the right to override any operation or default that touches it. These rights are architectural—available as a design property, not conditioned on workflow checkpoints or vendor permission. Paper 2 extends this commitment through per-mechanism governance shapes across intra-Self evolutionary dynamics. Paper 3 §4 carries the commitment across the inter-Self perimeter: the shared substrate constructed during a Full Aspect Integration event is itself substrate content, and its configuration is authored under joint authority of the participating Selves' governance rather than fixed by infrastructure.

Paper 3 Claim 5 (D0.05) states this at the claim level: configuration-as-substrate-content with recursive applicability is what Paper 1 §3.2's substrate-as-content commitment plus Paper 2 §8's per-mechanism governance shapes plus Paper 3 §4's Axis 2 commitment produce when applied to FAI events at the inter-Self perimeter. The cumulative configurable dimensions referenced forward through Paper 3 §§5–7 are themselves substrate content authored under joint authority across participating Selves' governance.

D1.22 formalizes the first of four sub-commitments that together constitute Claim 5: the enumeration of six specific dimensions at the FAI-event level, each independently governed as authored substrate content and each independently claimable as prior art. The note establishes that no dimension is hardwired by infrastructure, that each is independently inspectable and modifiable under the three rights, and that together they exhaust the FAI configuration space at the event level.

2. Statement of D1.22

D1.22. Every configurable dimension of the FAI protocol at the event level is authored substrate content under joint authority of the participating Selves' governance, individually subject to Paper 1 Claim 3's three rights (inspect, modify, override), and independently claimable as prior art. The six dimensions are: (1) sharing scope, (2) cardinality, (3) persistence policy, (4) cooperation/competition variant, (5) provenance carry-over depth at the perimeter, and (6) provenance preservation on internalization. Each dimension is independently governable. A system that governs any one of the six as authored substrate content partially satisfies Claim 5. A system that governs all six fully satisfies Claim 5 at the event-level configuration tier. The six together constitute the complete FAI configuration territory at this tier.

3. The six dimensions

Dimension 1 — Sharing scope

What it governs. Sharing scope specifies which aspects each participating Self contributes to the shared substrate for a given FAI event. An aspect is the exchange unit from Paper 2: the constituent DNA-layer content (orchestration substrates, behavior substrates, schemas, rules) and action-layer content (recorded task instances, outputs, lived experience) of a Self's cells. Not all aspects held by a participating Self need be contributed; governance determines which aspects to include in a given event. The sharing scope specification is the authored record of that determination.

Range. The range runs from minimal sharing (a single named aspect contributed by one Self, with all others excluded) to full sharing (all aspects of all participating Selves made available in the shared substrate). Intermediate specifications—sharing by category of aspect, by recency, by relevance to a named coordination goal, by explicit aspect identifier—are all within range. The governance decision may also specify asymmetric sharing, where different participating Selves contribute different subsets.

How it is authored as substrate content. The sharing scope specification is written into the FAI configuration substrate as an explicit, named record before or at the opening of the event. It identifies which aspects are in scope, under whose authority each inclusion decision was made, and what version of that specification is active. The specification is not inferred from participation or derived from prior events unless governance explicitly authors an inheritance rule.

Satisfaction of the three rights. Governance can inspect the sharing scope specification at any time—including mid-event—to confirm that the active record reflects current intent. Governance can modify the specification to include or exclude aspects, widen or narrow the scope, or substitute a different category rule. Governance can override any default or infrastructure-inferred scope that does not match the authored record.

Dimension 2 — Cardinality

What it governs. Cardinality specifies how many Selves participate in a single FAI event. The cardinality floor is two—one-Self self-derived combination is intra-Self territory per Paper 2, not an inter-Self FAI event. The cardinality ceiling is unbounded by architecture; it is a governance decision per event.

Range. The range runs from two (the minimal bilateral case) to any number of Selves whose participation is authorized by the configuration substrate. Cardinality may be fixed at event creation or left open within governance-specified bounds (minimum and maximum participants, or an open-ended participation window). The specification may also name particular Selves by identifier or name the class of Selves eligible to participate.

How it is authored as substrate content. The cardinality specification is an authored record in the FAI configuration substrate identifying the number or set of participating Selves and the authorization basis for that participant set. It is distinct from the operational list of Selves that actually join—the specification is the governance-authored intent; the operational list may diverge only within tolerances the specification itself defines.

Satisfaction of the three rights. Governance can inspect the cardinality specification to verify the authorized participant set at any time. Governance can modify it to add or remove authorized participants, change the minimum or maximum, or close enrollment. Governance can override any infrastructure default that would admit or exclude Selves without matching the authored specification.

Dimension 3 — Persistence policy

What it governs. Persistence policy specifies what remains of the shared substrate after the FAI event dissolves and where that remainder is held. Paper 3 §4 establishes that the shared substrate is a temporary construction across the inter-Self perimeter; dissolution is the default end state. Persistence policy governs what, if anything, survives dissolution and under what custody.

Range. The range runs from full dissolution (nothing of the shared substrate persists; all content either migrates to home substrates per the evolution feed or is discarded) to full audit record retention (a complete record of the shared substrate's content, provenance, and conflict history is preserved at a designated governance-accessible location). Intermediate positions include: retaining conflict records only; retaining the provenance chain of evolved content without retaining content itself; time-bounded retention with automatic expiry. The location of retained content—which Self's substrate carries the record, or whether a joint governance store holds it—is also a dimension of the persistence policy specification.

How it is authored as substrate content. The persistence policy is an authored record in the FAI configuration substrate specifying what survives dissolution, in what form, and under whose custody. It references the applicable retention authority and any expiry or audit rules that govern the retained record. The policy is authored before the event opens and is modifiable by governance during the event up to the moment of dissolution.

Satisfaction of the three rights. Governance can inspect the persistence policy at any time to confirm what will or did survive dissolution. Governance can modify the policy to extend, reduce, or redirect retention. Governance can override any platform default regarding post-dissolution retention that does not match the authored policy.

Dimension 4 — Cooperation/competition variant

What it governs. The FAI mechanism—shared substrate, aspect exchange, orchestration over contributed content—is a single architectural primitive. That primitive can operate under orchestration rule sets designed for cooperative coordination (Selves contributing toward a shared goal, with conflict resolution oriented toward synthesis) or competitive coordination (Selves contributing under conditions of measured opposition, with orchestration rules governing how competing contributions are evaluated or ranked). Dimension 4 specifies which variant's orchestration rule set governs a given FAI event.

Range. The range is, at minimum, the two named variants—cooperative and competitive—with any number of hybrid or composite rule sets within range if governance authors them. The variant specification is a selection among available orchestration rule sets, each of which must itself be authored substrate content under the three rights. Governance may also specify a transition rule: an event that begins under one variant's rule set and switches to another's upon a specified condition is within range if the transition condition and the applicable rule sets are both authored.

How it is authored as substrate content. The variant specification is an authored record in the FAI configuration substrate naming the active orchestration rule set and the governance authority that selected it. The architectural primitives—sharing scope, cardinality, persistence policy, provenance handling—are identical across variants. What differs is the orchestration rule set that governs how shared-substrate content is processed, how conflicts among contributed aspects are handled, and how evolved outputs are produced. Because the primitives are the same, a governance actor can change the variant specification without restructuring any other dimension.

Satisfaction of the three rights. Governance can inspect the variant specification to confirm which orchestration rule set is active. Governance can modify it to substitute a different rule set before or during an event, within any bounds the configuration substrate establishes. Governance can override any infrastructure inference about which rule set applies that does not match the authored specification. The variant selection is never an infrastructure default that governance cannot inspect or replace.

Dimension 5 — Provenance carry-over depth at the perimeter

What it governs. When a participating Self contributes an aspect to the shared substrate, that aspect carries provenance: a lineage chain documenting its authorship, DNA version history, governance authorization records, and any prior evolution events that shaped it. Dimension 5 specifies how much of that provenance history travels with the contributed aspect *into* the shared substrate. This is an ingress-boundary governance decision: it fires at the moment of contribution, before the aspect is active in the shared substrate.

Range. The range runs from minimal carry-over (only the aspect's immediate governance specifications—its current DNA version and the authorization for its contribution—travel into the shared substrate; prior lineage is not carried) to full carry-over (the complete lineage chain and DNA version history of the contributed aspect, including all prior evolution events, travel into the shared substrate). Intermediate specifications are within range: carry-over bounded to a named number of lineage steps; carry-over of selected provenance fields (authorship but not version history; conflict records but not lineage chain).

How it is authored as substrate content. The carry-over depth specification is an authored record in the FAI configuration substrate defining what provenance fields and what depth of lineage chain travel with contributed aspects. It may be specified uniformly for all aspects in the event or per-aspect-class. The specification is distinct from the provenance records themselves; it is the governance decision about how much of those records crosses the perimeter.

Satisfaction of the three rights. Governance can inspect the carry-over depth specification to confirm what provenance is entering the shared substrate with contributed aspects. Governance can modify it to expand or restrict the carry-over, or to specify different rules for different aspect classes. Governance can override any infrastructure default about what provenance migrates across the perimeter that does not match the authored specification.

Dimension 6 — Provenance preservation on internalization

What it governs. When an FAI event dissolves and participating Selves ingest evolved outputs into their home substrates via the four-locus evolution feed, those outputs carry provenance from the FAI event: records of which aspects contributed to them, which conflicts were encountered and resolved, which orchestration rule set governed, and what the FAI event's own governance specifications were. Dimension 6 specifies how much of that FAI-origin provenance is preserved when evolved content moves *into* the home substrate. This is an egress-boundary governance decision: it fires at the moment of internalization, after the FAI event's content has been processed and is ready for home-substrate ingestion.

Range. The range runs from FAI-event reference only (provenance in the home substrate notes the FAI event as source by identifier, with no further detail about contributing aspects or event-internal history) to full provenance carry (the complete contributing-aspect lineage chain, the FAI event's conflict records, and the event's full configuration provenance are preserved in the home substrate records alongside the evolved content). Intermediate specifications include: preserving contributing-aspect identifiers without full lineage; preserving conflict records but not lineage chains; preserving provenance to a bounded depth beyond the FAI-event reference.

How it is authored as substrate content. The internalization provenance specification is an authored record in the FAI configuration substrate (and, correspondingly, in each participating Self's home-substrate configuration) defining what FAI-origin provenance fields are preserved upon ingestion. Because this decision concerns the home substrate, it is a joint governance decision: the configuration is authored under the home Self's authority for home-substrate effects, within whatever joint-authority bounds the FAI event's configuration substrate establishes. The specification is independent of Dimension 5—the two dimensions govern different moments and different substrate boundaries.

Satisfaction of the three rights. Governance can inspect the internalization provenance specification in both the FAI configuration substrate and the relevant home substrate to confirm what provenance will or did carry into the home records. Governance can modify the specification to extend, restrict, or redirect provenance preservation before the event's evolution feed executes. Governance can override any infrastructure default about provenance retention at the home substrate boundary that does not match the authored specification.

4. Severability

Each of the six dimensions enumerated above is independently governable and independently claimable as prior art. The severability commitment has the following consequences.

Independent prior-art coverage. An adversary cannot claim novelty for any single dimension because D1.22 covers all six. A system that authors only the persistence policy as substrate content while leaving all other dimensions as infrastructure defaults or vendor determinations has still entered prior-art territory on Dimension 3. A system that authors only provenance carry-over depth has still entered prior-art territory on Dimension 5.

Partial implementation falls within coverage. A system that governs some but not all six dimensions through authored substrate content partially satisfies Claim 5. Partial satisfaction is not a failure to infringe the prior-art territory; it demonstrates that the system has adopted the architectural posture for those dimensions it does govern.

Full satisfaction requires all six. A system that governs all six dimensions through authored substrate content under joint governance authority, with each dimension subject to the three rights at all times, fully satisfies Claim 5 at the event-level configuration tier. D1.23’s recursive property and D1.25’s joint-authority requirement further constrain full satisfaction.

Dimensions are architecturally independent. No dimension is derived from or dependent on the others in the sense that governing one forces a particular specification for another. Sharing scope and cardinality are independent governance decisions. Persistence policy and provenance handling do not interact except through governance-authored rules that explicitly connect them. Cooperation/competition variant is a selection over orchestration rule sets that operates over whatever sharing scope and cardinality governance has specified.

5. Operational test

For any system claiming to instantiate FAI or inter-Self coordination over shared substrate, the following test checks D1.22 compliance. The test applies to each FAI event independently, since the six dimensions may be specified differently across events.

For a given FAI event, an observer with governance access asks:

1. **Sharing scope.** Is there an authored record—created or approved by governance authority, not inferred by infrastructure—specifying which aspects each participating Self contributes? Can governance inspect that record now, without requesting platform access? Can governance modify it before the event opens or before the contributing Self’s aspects are merged into the shared substrate?

2. **Cardinality.** Is there an authored record specifying the authorized participant set for this event? Is that record distinct from the operational execution log that records who in fact joined? Can governance inspect, modify, or override it independent of the event’s execution state?

3. **Persistence policy.** Is there an authored record specifying what will survive dissolution, in what form, and under whose custody? Does that record identify the applicable retention authority? Can governance inspect and modify it before dissolution?

4. **Cooperation/competition variant.** Is there an authored record naming the active orchestration rule set and the governance authority that selected it? Is that record distinguishable from the rule set itself (the selection record is the dimension; the rule set is what it selects)? Can governance replace the selection without restructuring any other dimension’s specification?

5. **Provenance carry-over depth.** Is there an authored record specifying what provenance fields and what lineage depth travel with contributed aspects across the perimeter into the shared substrate? Does that record apply at the moment of contribution (ingress boundary) rather than at dissolution or internalization? Can

governance inspect and modify it per-aspect-class or uniformly?

6. **Provenance preservation on internalization.** Is there an authored record—in the FAI configuration substrate and/or the relevant home-substrate configuration—specifying what FAI-origin provenance is preserved upon ingestion into the home substrate? Is that record independent of the carry-over depth record (Dimension 5), governing a distinct moment at the egress boundary? Can governance inspect and modify it before the evolution feed executes?

If the answer to all questions across all six dimensions is yes, the system satisfies D1.22. If any dimension lacks an authored record (that is, the specification is an infrastructure default, a vendor determination, or an emergent consequence of execution rather than a governance-authored artifact), that dimension falls outside D1.22 compliance and outside full Claim 5 satisfaction at the event-level tier.

6. Forward: remaining Claim 5 sub-commitments

D1.22 enumerates the six event-level dimensions. Three further sub-commitments complete the Claim 5 set:

D1.23 — Configuration-of-configuration. The configuration substrate carrying the six dimensions is itself substrate content under joint authority. Configuring the configuration substrate is also a substrate-content act under the three rights. The recursion is an architectural property of the substrate-content commitment, not a fresh meta-governance layer, and it bottoms out at human-authored governance authority per Paper 1 §3.3.

D1.24 — Recursive applicability across all Paper 3 claims. The seventeen cumulative configurable dimensions across Paper 3 §§5–7—six at the FAI-event level (D1.22), six at the conflict-handling level, five at the evolution-feed level—all carry forward as substrate content under joint authority per Claim 5's recursive applicability commitment. D1.24 formalizes this cross-claim extension.

D1.25 — Joint authority over the configuration substrate. The FAI configuration substrate is authored under joint authority of the participating Selves' governance, not under the unilateral authority of any one participant. D1.25 formalizes the multi-party governance structure that joint authorship requires, including the commitment that no single participating Self can unilaterally modify the configuration substrate in ways that bind other participants without their governance authorization.

Source

Li, W. (April 2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. (Paper 3 in the CKS theory series.) Independent publication. §8 (Claim 5 — Configuration as substrate content with recursive applicability); §5 (configurable dimensions at the FAI-event level).

Li, W. (April 2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. (Paper 1 in the CKS theory series.) Independent publication. §3.2 (substrate-as-content); §3.3 (authority-vs-labor distinction; three rights).

Li, W. (April 2026). *The Instinct/Reasoning Separation Outside the Model*. (Paper 2 in the CKS theory series.) Independent publication. §8 (per-mechanism governance shapes).